FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SEMESTER VII
SEMINAR (CSQ413)

ACADEMIC YEAR : 2026 – 2027 **BATCH :** CSD

Seminar Title :	**Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection**

Student Name :	**Joseph D Poonoly (Roll No: 36, Slot No: 27)**

Type of Seminar :

[✓] **Review Seminar** (presenting a journal/conference/survey paper)

[ ] Technical / Concept Seminar (explaining an algorithm, technology, architecture, protocol, etc.)

[ ] Application Seminar (showing how a technology is applied in real-world systems)

[ ] Product / Tool Seminar (overview of a software/hardware product, platform, or framework)

Project Mapping

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
x	x	x	x	x	x	x	x	x	x	x	x	x

Mapping Justification

S No	PO/PSO	Justification
1	PO1	Apply computer science and engineering knowledge to understand distributed machine learning, convolutional neural networks, federated averaging, and gradient optimization.
2	PO2	Analyze semiconductor wafer defect patterns (center, donut, edge-ring, mixed types) and examine the privacy limitations of centralized training across proprietary fabrication facilities.
3	PO3	Understand the design of a privacy-preserving federated inspection framework combining ResNet152 backbones, FedAvg aggregation, differential privacy, and explainable AI.
4	PO4	Examine the benchmark dataset (MixedWM38 with 38,000+ images), exploratory data analysis, PCA dimensionality reduction, and evaluation metrics across 10 communication rounds.

S No	PO/PSO	Justification
5	PO5	Understand the use of PyTorch, differential privacy libraries, Grad-CAM, LIME, and t-SNE visualization tools in implementing the semiconductor fault detection framework.
6	PO6	Recognize how automated defect detection in semiconductor manufacturing enhances hardware quality, device reliability, and cost efficiency in automotive and medical electronics.
7	PO7	Understand the importance of data confidentiality in industrial AI, safeguarding proprietary wafer map designs through differential privacy without exposing IP.
8	PO8	Develop the ability to independently study peer-reviewed IEEE Access research, organize technical findings, and collaborate in preparing and presenting the seminar.
9	PO9	Improve technical communication by clearly explaining federated aggregation formulas, Gaussian noise injection, Grad-CAM heatmaps, and performance benchmarks.
10	PO10	Understand resource planning, communication round budgets, and computational requirements when deploying federated learning models across distributed manufacturing fabs.
11	PO11	Develop the ability to explore emerging frontiers in edge federated learning, MLOps CI/CD pipelines, and quantum-accelerated semiconductor inspection.
12	PSO1	Apply deep learning and image processing principles to model semiconductor wafer map patterns and design explainable visual feature attribution mechanisms.
13	PSO2	Understand the integration of privacy-preserving decentralized algorithms with post-hoc interpretability models for reliable industrial AI engineering.

Signature (Seminar Coordinator) : _____